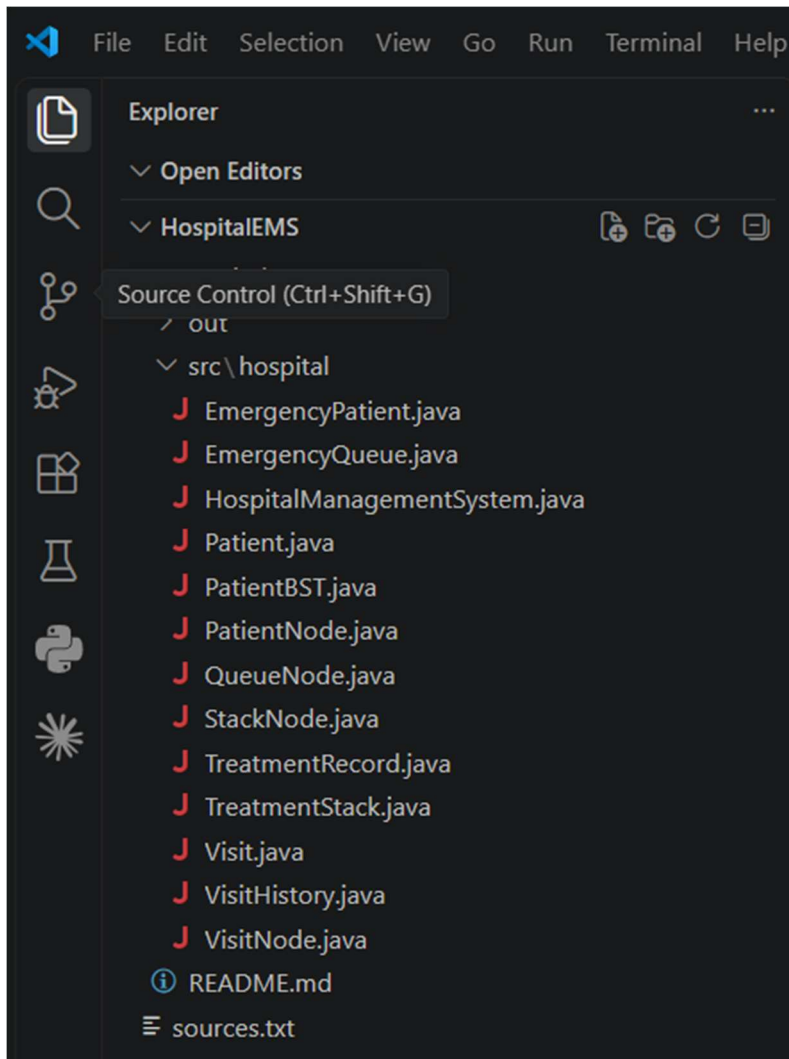


Screenshots of program output

23DA2-0507

Screenshot 1: Project structure and main menu

Purpose: This proves that the project has an organized Java structure and that the system runs.



```

● PS C:\Users\Admin\Desktop\HospitaleMS> cd HospitaleMS
○ PS C:\Users\Admin\Desktop\HospitaleMS\HospitaleMS> javac -d out src\hospital\*.java
>> java -cp out hospital.HospitalManagementSystem
=====
Mini Hospital Emergency Management System
=====

----- MAIN MENU -----
Patient Records (BST)
1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)
Emergency Queue
5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue
Treatment History (Stack)
8. Display treatment history
9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit
-----
Enter your choice: 

```

Screenshot 2: Patient registration

Purpose: This demonstrates that patient records can be registered and stored in the BST.

```

----- MAIN MENU -----
Patient Records (BST)
1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)
Emergency Queue
5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue
Treatment History (Stack)
8. Display treatment history
9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit
-----

Enter your choice: 1
Enter Patient ID: 104
Enter Patient Name: Nethmi Ayodya
Enter Age: 22
Enter Contact Number: 0711111111
Enter Medical Condition: Asthma
Patient registered successfully.

```

```

----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
 10. Add visit to a patient's history
 11. Remove a visit from a patient's history
 12. Search for a visit in a patient's history
 13. Display a patient's visit history
  0. Exit
-----

Enter your choice: 1
Enter Patient ID: 102
Enter Patient Name: Sujith Kumara
Enter Age: 35
Enter Contact Number: 0722222222
Enter Medical Condition: Fever
Patient registered successfully.

```

```

----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
 10. Add visit to a patient's history
 11. Remove a visit from a patient's history
 12. Search for a visit in a patient's history
 13. Display a patient's visit history
  0. Exit
-----

Enter your choice: 1
Enter Patient ID: 101
Enter Patient Name: Sanduni Piyumi
Enter Age: 28
Enter Contact Number: 0733333333
Enter Medical Condition: Chest pain
Patient registered successfully.

```

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 1
Enter Patient ID: 103
Enter Patient Name: Mithun Lakmal
Enter Age: 42
Enter Contact Number: 0744444444
Enter Medical Condition: Accident injury
Patient registered successfully.
```

Screenshot 3: BST in-order traversal

Purpose: This is essential evidence for the Binary Search Tree. It proves that the patients are displayed in ascending Patient ID order.

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 4
---- Patients (ascending Patient ID) ----
ID: 101 | Name: Sanduni Piyumi | Age: 28 | Contact: 0733333333 | Condition: Chest pain
ID: 102 | Name: Sujith Kumara | Age: 35 | Contact: 0722222222 | Condition: Fever
ID: 103 | Name: Mithun Lakmal | Age: 42 | Contact: 0744444444 | Condition: Accident injury
ID: 104 | Name: Nethmi Ayodya | Age: 22 | Contact: 0711111111 | Condition: Asthma
```

Screenshot 4: BST search and deletion

Purpose: This proves the BST search and delete operations.

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 2
Enter Patient ID to search: 103
Patient found:
ID: 103 | Name: Mithun Lakmal | Age: 42 | Contact: 0744444444 | Condition: Accident injury
```

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 3
Enter Patient ID to delete: 102
Patient deleted successfully.
```

```

----- MAIN MENU -----
Patient Records (BST)
1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)
Emergency Queue
5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue
Treatment History (Stack)
8. Display treatment history
9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit
-----
Enter your choice: 4
---- Patients (ascending Patient ID) ----
ID: 101 | Name: Sanduni Piyumi | Age: 28 | Contact: 0733333333 | Condition: Chest pain
ID: 103 | Name: Mithun Lakmal | Age: 42 | Contact: 0744444444 | Condition: Accident injury
ID: 104 | Name: Nethmi Ayodya | Age: 22 | Contact: 0711111111 | Condition: Asthma

```

Screenshot 5: Emergency queue and FIFO behavior

Purpose: This proves that the emergency queue follows FIFO: First-In, First-Out.

Because patient 102 was deleted in the previous step, use patient IDs 101, 103, and 104.

```

----- MAIN MENU -----
Patient Records (BST)
1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)
Emergency Queue
5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue
Treatment History (Stack)
8. Display treatment history
9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
Please enter a valid whole number.
Enter your choice: 5
Enter Patient ID: 101
Enter reason for emergency visit: Chest Pain
Added to emergency queue: ID: 101 | Name: Sanduni Piyumi | Reason: Chest Pain

```

----- MAIN MENU -----

Patient Records (BST)

1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)

Emergency Queue

5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue

Treatment History (Stack)

8. Display treatment history
9. Undo / remove most recent treatment record (pop)

Patient Visit History (Linked List)

10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit

Enter your choice: 5

Enter Patient ID: 103

Enter reason for emergency visit: Accident injury

Added to emergency queue: ID: 103 | Name: Mithun Lakmal | Reason: Accident injury

----- MAIN MENU -----

Patient Records (BST)

1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)

Emergency Queue

5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue

Treatment History (Stack)

8. Display treatment history
9. Undo / remove most recent treatment record (pop)

Patient Visit History (Linked List)

10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit

Enter your choice: 5

Enter Patient ID: 104

Enter reason for emergency visit: Asthma attack

Added to emergency queue: ID: 104 | Name: Nethmi Ayodya | Reason: Asthma attack

```

----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit

```

Enter your choice: 7

---- Patients waiting (front -> rear) ----

1. ID: 101	Name: Sanduni Piyumi	Reason: Chest Pain
2. ID: 103	Name: Mithun Lakmal	Reason: Accident injury
3. ID: 104	Name: Nethmi Ayodya	Reason: Asthma attack

Screenshot 6: Queue dequeue and treatment completion

Purpose: In HospitalEMS version, option 6 performs two related actions: it dequeues the next emergency patient and records the completed treatment in the stack.

```

----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit

```

Enter your choice: 6

Now treating: ID: 101 | Name: Sanduni Piyumi | Reason: Chest Pain

Enter treatment given: Emergency chest-pain assessment and medication

Enter completion date/time (e.g. 2026-09-02 10:30): 2026/09/05 08:30

Treatment record pushed to history: PatientID: 101 | Name: Sanduni Piyumi | Treatment: Emergency chest-pain assessment and medication | Completed: 2026/09/05 08:30

Screenshot 7: Treatment stack and LIFO behavior

Purpose: This proves that completed treatments are stored using a Stack and follow LIFO: Last-In, First-Out.

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to
Please enter a valid whole number.
Enter your choice: 6   Injury examination and first aid
Enter completion date/time (e.g. 2026-09-02 10:30): 2026/09/03 09:30
Treatment record pushed to history: PatientID: 103   | Name: Mithun Lakmal   | Treatment: Injury
examination and first aid | Completed: 2026/09/03 09:30
```

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 8
---- Treatment history (most recent first) ----
1. PatientID: 103   | Name: Mithun Lakmal   | Treatment: Injury examination and first aid |
Completed: 2026/09/03 09:30
2. PatientID: 101   | Name: Sanduni Piyumi   | Treatment: Emergency chest-pain assessment and
medication | Completed: 2026/09/05 08:30
```

----- MAIN MENU -----

Patient Records (BST)

1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)

Emergency Queue

5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue

Treatment History (Stack)

8. Display treatment history
9. Undo / remove most recent treatment record (pop)

Patient Visit History (Linked List)

10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit

Enter your choice: 9

Removed most recent treatment record: PatientID: 103 | Name: Mithun Lakmal | Treatment:
Injury examination and first aid | Completed: 2026/09/03 09:30

Screenshot 8: Add patient visits to the linked list

Purpose: This proves that visits can be added to a patient's Singly Linked List.

Use patient ID 101, Sanduni Piyumi.

----- MAIN MENU -----

Patient Records (BST)

1. Register new patient
2. Search patient by ID
3. Delete patient by ID
4. Display all patients (in-order)

Emergency Queue

5. Add patient to emergency queue
6. Call next patient for treatment (dequeue)
7. Display waiting queue

Treatment History (Stack)

8. Display treatment history
9. Undo / remove most recent treatment record (pop)

Patient Visit History (Linked List)

10. Add visit to a patient's history
11. Remove a visit from a patient's history
12. Search for a visit in a patient's history
13. Display a patient's visit history
0. Exit

Enter your choice: 10

Enter Patient ID: 101

Enter visit date (e.g. 2026-01-15): 2026-08-20

Enter doctor name: Dr.Mohammed

Enter diagnosis: Chest pain

Enter treatment given: Medical examination and prescribed medication

Visit added with Visit ID 1000

```

----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 10
Enter Patient ID: 101
Enter visit date (e.g. 2026-01-15): 2026-09-05
Enter doctor name: Dr.Kumar
Enter diagnosis: Follow-up consultation
Enter treatment given: Continued observation and review
Visit added with Visit ID 1001

```

Screenshot 9: Display visit history

Purpose: This proves that the linked list stores and displays multiple visits for one patient.

```

----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 13
Enter Patient ID: 101
Visit history for Sanduni Piyumi (ID 101):
  - VisitID: 1000 | Date: 2026-08-20 | Doctor: Dr.Mohammed | Diagnosis: Chest pain
    | Treatment: Medical examination and prescribed medication
  - VisitID: 1001 | Date: 2026-09-05 | Doctor: Dr.Kumar | Diagnosis: Follow-up
    consultation | Treatment: Continued observation and review

```

Screenshot 10: Search and remove a visit

Purpose: This proves the linked-list search and remove operations.

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next pati
Please enter a valid whole number.
Enter your choice: 12
Enter Patient ID: 101
Enter Visit ID to search: 1000
Found visit: VisitID: 1000 | Date: 2026-08-20 | Doctor: Dr.Mohammed | Diagnosis: Chest pain
| Treatment: Medical examination and prescribed medication
```

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
  10. Add visit to a patient's history
  11. Remove a visit from a patient's history
  12. Search for a visit in a patient's history
  13. Display a patient's visit history
  0. Exit
-----
Enter your choice: 11
Enter Patient ID: 101
Enter Visit ID to remove: 1000
Visit removed.
```

Screenshot 11: Empty queue and empty stack handling

Purpose: This demonstrates appropriate handling of empty data structures, which is specifically required by the assignment.

```
----- MAIN MENU -----ry
Patient Records (BST)t's visit history
  1. Register new patient
  2. Search patient by ID-----
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
 10. Add visit to a patient's history
 11. Remove a visit from a patient's history
 12. Search for a visit in a patient's history
 13. Display a patient's visit history
  0. Exit

-----
Enter your choice: 7
---- Patients waiting (front -> rear) ----
1. ID: 104   | Name: Nethmi Ayodya   | Reason: Asthma attack
```

```
----- MAIN MENU -----
Patient Records (BST)
  1. Register new patient
  2. Search patient by ID
  3. Delete patient by ID
  4. Display all patients (in-order)
Emergency Queue
  5. Add patient to emergency queue
  6. Call next patient for treatment (dequeue)
  7. Display waiting queue
Treatment History (Stack)
  8. Display treatment history
  9. Undo / remove most recent treatment record (pop)
Patient Visit History (Linked List)
 10. Add visit to a patient's history
 11. Remove a visit from a patient's history
 12. Search for a visit in a patient's history
 13. Display a patient's visit history
  0. Exit

-----
Enter your choice: 9
Treatment history is empty. Nothing to pop.
```